CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Fifth Semester of B. Pharm. Examination
University Theory Examination March 2019
PH319/PH334 Pharmaceutical Microbiology and Biotechnology

Date: 14/03/2019, Thursday **Time:** 01:30 p.m. to 04:30 p.m. **Maximum Marks:** 80

Instructions:

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat labeled sketches wherever necessary.**

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. Peptidoglycan accounts for __________ of the dry weight of cell wall in many gram positive bacteria.
[A] 50% or more
[B] About 10%
[C] 11% ± 0.22%
[D] About 20%

2. The common word for bacteria which are spherical in shape is _______.
[A] Cocci
[B] Bacilli
[C] Spirilla
[D] Pleomorphic

3. In the three domain system of classification, the traditional bacteria are placed in the ________.
[A] eukarya
[B] archaea
[C] eubacteria
[D] archeobacteria

4. What crucial feature of a penicillin is involved in its mechanism of action?
[A] Carboxylic Acid
[B] Beta Lactum Ring
[C] Acyl Side Chain
[D] Thiazolidine Ring

5. Ultraviolet light can cause the formation of double bonds between adjacent pyrimidines, resulting in a formation called a _____.
[A] pyrimidine mispairing
[B] pyrimidine-pyrimidine pairing
[C] pyrimidine dimer
[D] pyrimidine pseudopairing

6. Which of the following is the best description of a gene?
[A] A molecule of DNA
[B] A nucleotide
[C] A sequence of DNA nucleotides that codes for a single protein
[D] A sequence of amino acids in a protein

7. The time required to kill 90% of the microorganisms in a sample at a specific temperature is the __________.
[A] Decimal Reduction Time
[B] Thermal Death Point
[C] F Value
[D] D Value

8. The action of alcohol during Gram-staining is to _____________.
[A] allow the color
[B] add the color
[C] decolorize the cells
[D] intensify the color

9. Temperature required for pasteurization is ____________.
[A] Above 150 °C
[B] Below 100 °C
[C] 130 °C
[D] 110 °C

10. Which of the following is ionizing radiation?
[A] UV rays
[B] IR
[C] γ-rays
[D] Visible light

11. The condition required for autoclave is ____________.
[A] 121 °C temperature and 15 lbs. pressure for 15 minutes
[B] 100 °C temperature and 30 lbs. pressure for 30 minutes
[C] 150 °C temperature for 60 minutes
[D] 130 °C temperature for 120 minutes

12. In the industrial production of streptomycin, the secondary metabolite or byproducts is ______.
[A] Vitamin B_{12}
[B] Vitamin C
[C] Vitamin B_6
[D] Ethanol

13. To differentiate lactose and non-lactose fermenters, the medium used is ________.
[A] Mac Conkey's medium
[B] Stuart's medium
[C] Sugar medium
[D] Citrate medium

14. The total magnification of a microscope is equal to ___________.
[A] the magnification of the objective lens × the number of eye pieces
[B] the magnification of the objective × the number of ocular
[C] the total of all the objectives
the price of the microscope

15. A membrane usually found to be quite suitable for sterility testing essentially bears a nominal pore size not more than _______ micrometers.

[A] 0.45
[B] 0.22
[C] 0.35
[D] 0.50

16. Which one of the following methods is not used for the total bacterial count study?

[A] Filtration method
[B] Pour plate method
[C] Spread plate method
[D] Hier plate method

17. Which of the following is bactericidal?

[A] Membrane Filtration
[B] Ionizing Radiation
[C] Freeze Drying
[D] Deep Freezing

18. If a 1:600 dilution of a test compound kills a standard population of Staphylococcus aureus in 10 minutes but not 7.5 minutes while a 1:60 dilution of phenol kills the population in the same time, what is the phenol coefficient of the test compound?

[A] 1
[B] 5
[C] 10
[D] 50

19. In Bacterial transforamtion the bacteria takes up and express foreign DNA usually in the form of a________?

[A] Plasmid
[B] Enzyme
[C] Cofactor
[D] RNA

20. Clinical thermometers are sterilized, before its use, using _________.

[A] Phenol
[B] Alcohol
[C] Methanol
[D] Formaldehyde

SECTION – II

Q 2	Attempt any **<u>TWO</u>** of the following	
A	Write a note on mechanism of action of phenol, halogens and heavy metals for their disinfectant property.	**05**
B	Suggest the appropriate method/s of sterilization for 0.9 %W/V NaCl Large volume parenteral, Chloramphenicol Eye Drops, Surgical sutures, Neomycin sulphate powder and Testosterone IM injection.	**05**
C	Desctribe the use of McConkey's medium for isolation of bacteria.	**05**

Q 3	Attempt any **TWO** of the following		
A	Draw a sketch of an autoclave. Explain the principle of working for an autoclave. Enlist the various applications of an autoclave.	**05**
B	Describe the direct inoculation method used for performing the 'tests for sterility'.	**05**
C	Describe the microbiological assay for vitamin B_{12}.	**05**

SECTION – III

Q 4	Attempt any **FOUR** of the following	
A	Define : Pharmaceutical Microbiology. Write its scopes and applications in detail.	**05**
B	Write a note on classification of bacteria.	**05**
C	Define: Sterilization. Explain in detail various factors affecting on sterilization.	**05**
D	Define: Disinfection. What is Dynamic of Disinfection?	**05**
E	What is sterility test? Explain in detail sterility tests for aqueous solutions.	**05**
F	What is zone of inhibition? Explain cup-plate method for microbiological assay of antibiotics.	**05**
Q 5	Attempt any **FOUR** of the following	
A	Write a note on – Growth curve of bacteria. List out factors affecting growth of micro-organisms.	**05**
B	What is protoplast fusion? Describe various methods utilized for protoplast fusion.	**05**
C	Describe various techniques used for enzyme immobilization.	**05**
D	Define: Fermentation. Draw neat and clean diagram of fermentor and explain its various parts in detail.	**05**
E	What is microbial biotransformation? Explain various types of microbial biotransformation.	**05**
F	Define: Recombinant DNA technology. Write its applications in detail.	**05**